Diabetes Prediction

Using Feature Grouping

24/April/2026

Original Dataset: 21 Features

- HighBP
- HighChol
- CholCheck
- Smoker
- Stroke
- HeartDiseaseorAttack
- PhysActivity
- Fruits
- Veggies
- HvyAlcoholConsump
- AnyHealthcare
- NoDocbcCost
- DiffWalk

- GenHlth (1–5)
- Age (1–13)
- Education (1–6)
- Income (1–8)

- BMI
- MentHlth (0–30 days)
- PhysHlth (0–30 days)

Target Variable

- Diabetes_binary (0/1)

Target: Diabetes_binary → 0 = No Diabetes / Pre-Diabetes | 1 = Diabetes

Three Big Groups

Demographic

Age · Gender · Education · Income

HEALTH STATUS

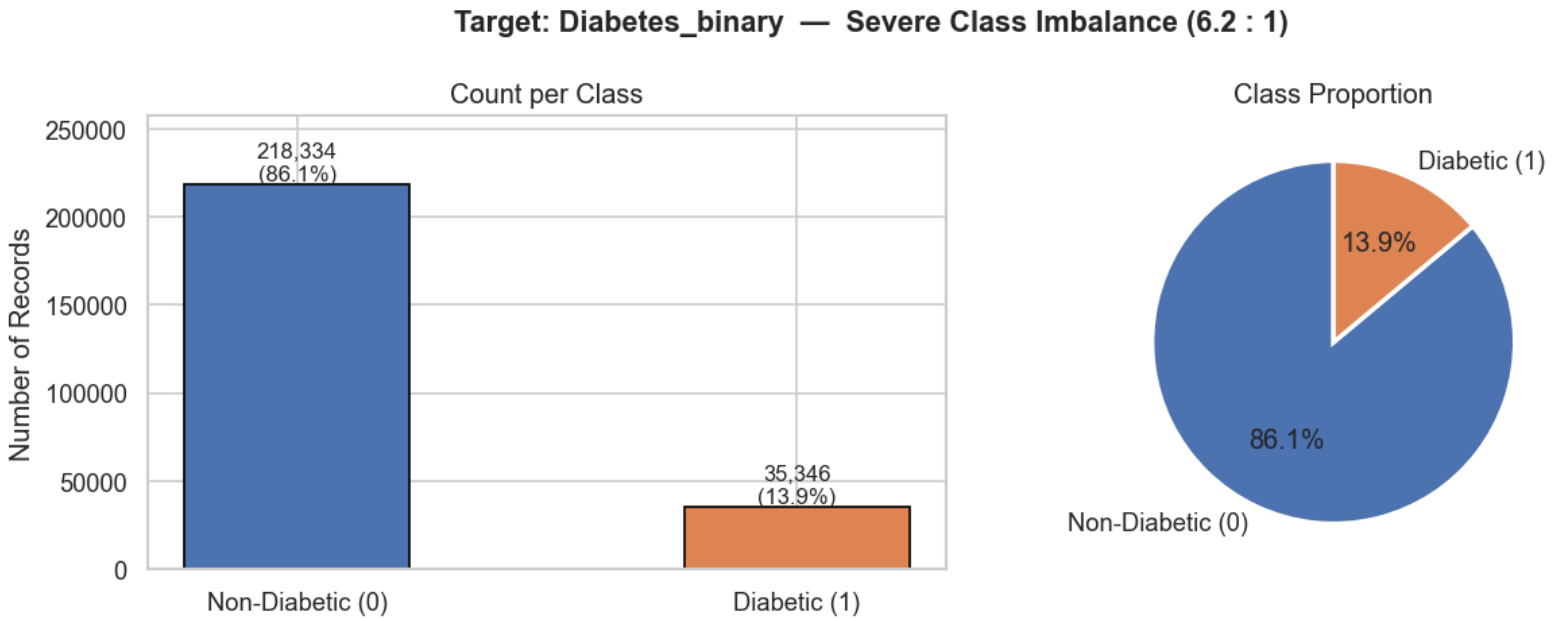
Blood pressure · Cholesterol · Stroke · Heart disease · General health · Physical & Mental health

Lifestyle & Healthcare Access

Smoking · Drinking · Exercise · Diet · Insurance · Cost issue · Doctor check

EDA — Class Imbalance

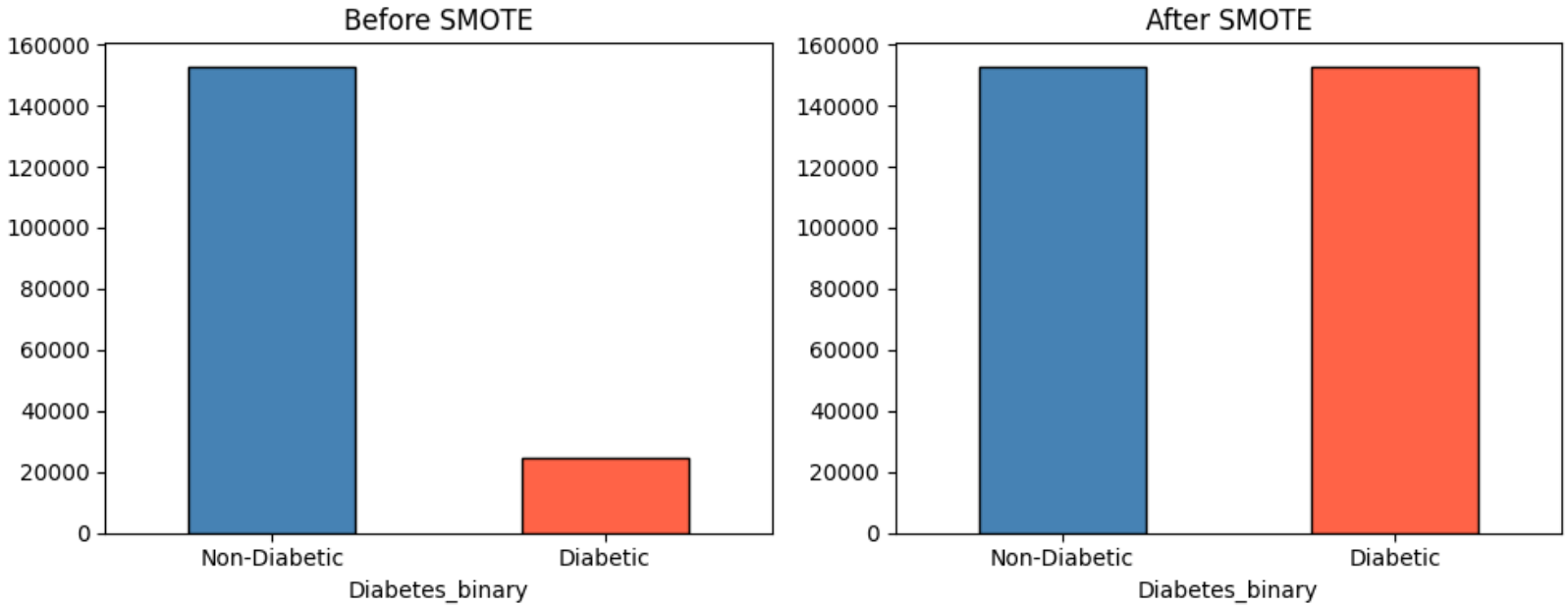
Data Overview (Imbalance)



The dataset is imbalance Use SMOTE to balance training data

EDA — Class Imbalance

Data Overview (Imbalance)

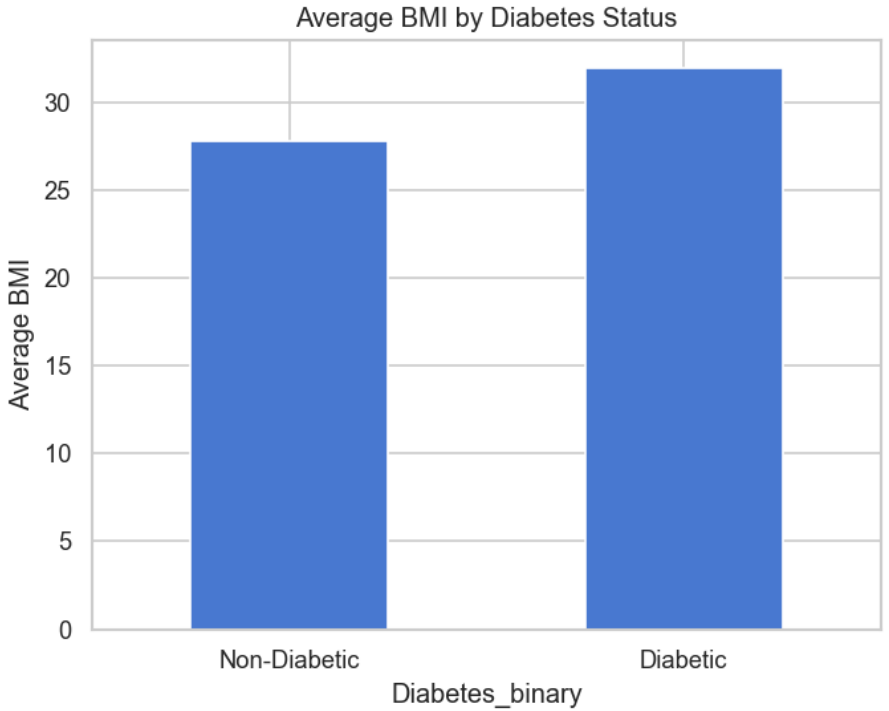


EDA – Key Risk Factors

BMI vs Diabetes

```
# average BMI
avg_bmi = df.groupby('Diabetes_binary')['BMI'].mean()

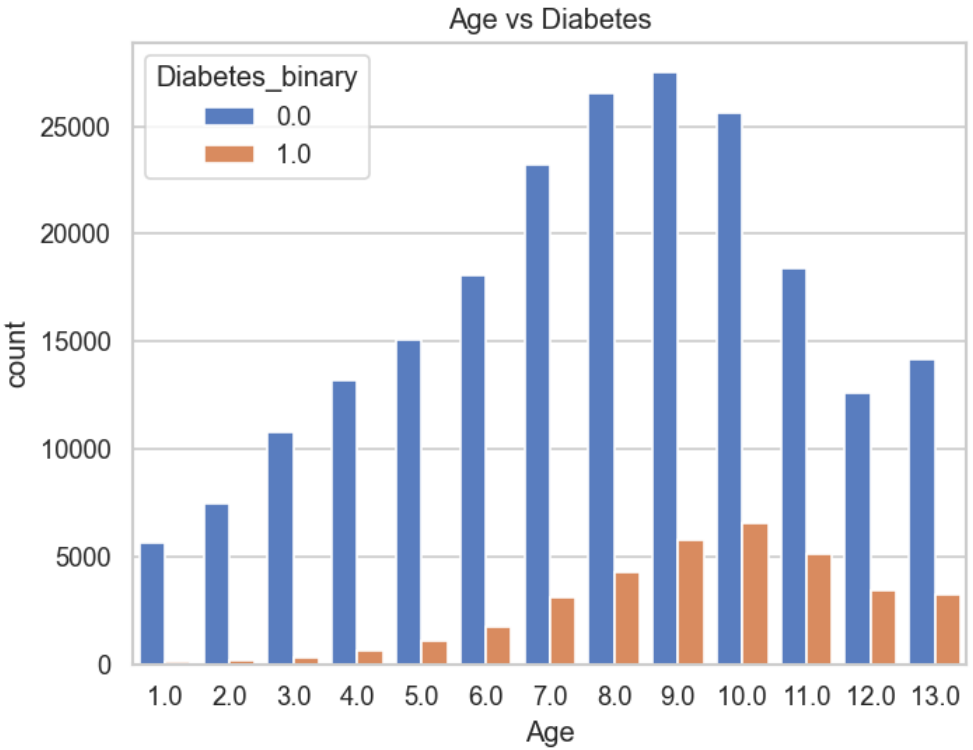
# plot
avg_bmi.plot(kind='bar')
plt.xticks([0,1], ['Non-Diabetic', 'Diabetic'], rotation=0)
plt.ylabel("Average BMI")
plt.title("Average BMI by Diabetes Status")
plt.show()
```



EDA – Key Risk Factors

Age VS Diabetes

```
plt.figure()
sns.countplot(x='Age',
hue='Diabetes_binary', data=df)
plt.title("Age vs Diabetes")
plt.show()
```



THANK YOU

AI-Powered Diabetes Risk Prediction

Royal University of Phnom Penh

Faculty of Engineering
Department of Data Science and Engineering

Explainable AI for Diabetes Risk Prediction and Actionable Insights

Extended Output Design

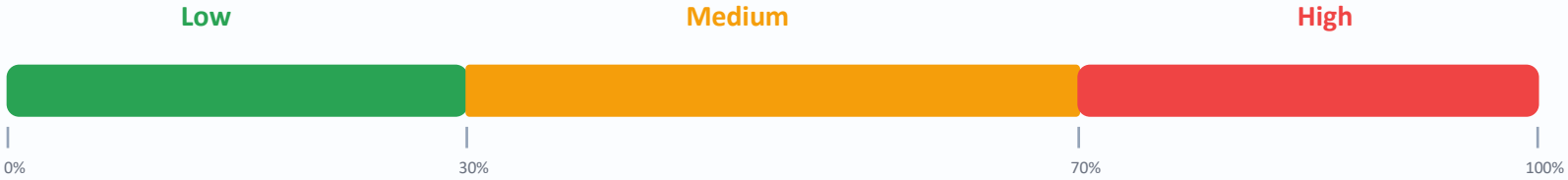


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Presenter: NHOR Povketya

06/May/2026

Output 1: Probability + Risk Level

Convert model probability into human-readable risk categories



Example A: 20%

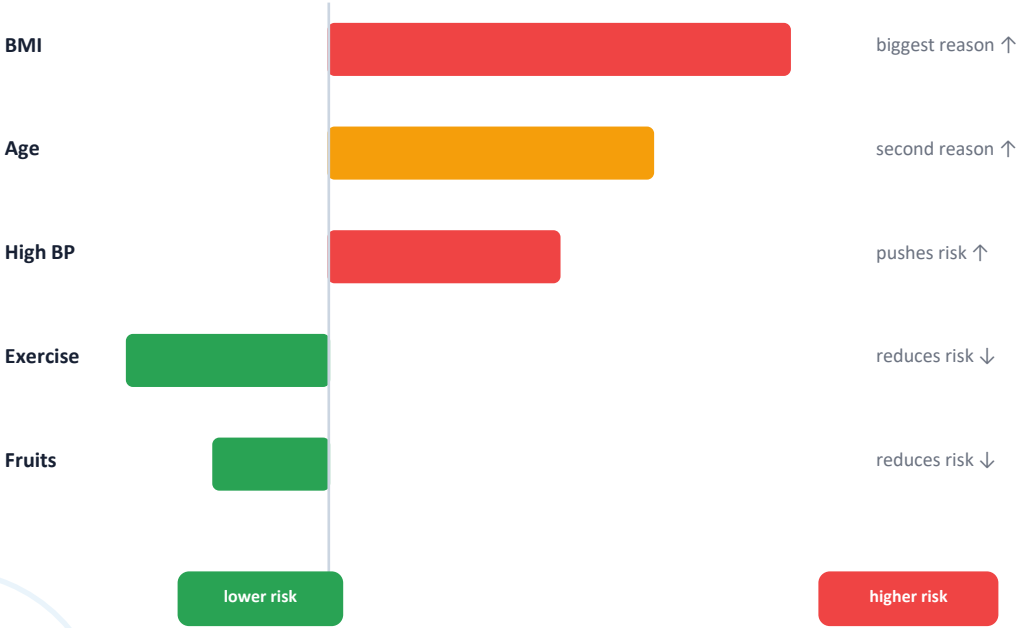
Risk level: Low

Example B: 78%

Risk level: High

Output 2: Feature Contribution

Patient risk explanation



Simple explanation

"This patient is high risk mainly because BMI and Age push the prediction upward. Physical activity helps reduce the risk."

Output 3: Group Insight

1

Demographic

Age, Sex, Education, Income

Who the patient is

2

Health Status

BMI, HighBP, HighChol, GenHlth, Heart disease

Current health condition

3

Lifestyle & Healthcare Access

Physical activity, fruits, veggies, smoking, healthcare access

Daily behavior and access

Example insight: “Health Status is the strongest driver because BMI and high blood pressure increase the prediction.”

Output 4: Explain + Suggest Improvement

Model Output

78% High Risk

→

Why

BMI + Age increase risk
Exercise reduces risk

→

Improve

More activity
Weight management
Regular check-up

Important: the system supports prevention and education. It should not claim to replace doctors or diagnose the patient.

Complete Extended Output Design

Patient Risk Report

78% High Risk

Probability	Risk score = 78% → High
Feature Contribution	BMI and Age increase risk; Physical Activity reduces risk
Group Insight	Health Status is the main driver; Lifestyle is modifiable
Recommendation	Increase activity, monitor BMI, check blood pressure / cholesterol

This output is easier for users because it answers: How risky? Why? Which group? What can be improved?

Actionable Recommendation: What does Gemini receive?

System Output

Prediction result
78% · High Risk

Patient profile
Age, BMI, BP, activity

SHAP explanation
Top factors: BMI, Age

Safety rule
Education only, not diagnosis

Structured Input Sent to Gemini

```
{
  "risk_probability": 0.78,
  "risk_level": "High",
  "patient_profile": {
    "Age": "58",
    "BMI": 34.7,
    "HighBP": 1,
    "HighChol": 1,
    "PhysActivity": 0
  },
  "top_risk_factors": ["BMI", "Age", "HighBP"],
  "protective_factors": ["Physical Activity"],
  "instruction": "Provide simple lifestyle
education only. Do not diagnose."
}
```

Important: Gemini explains and recommends. The ML model already made the prediction.

Gemini Output

1. Simple explanation: why risk is high
2. 3–5 actions: exercise, diet, BMI tracking
3. Safe reminder: consult health professional
4. Friendly wording for patient/user

"The AI recommendation uses three inputs: the risk score, patient health data, and SHAP reasons. Gemini only translates them into understandable lifestyle advice."

Final Output Example

Patient Result

78%

High Risk



Predicted diabetes risk probability

Main Reasons

1. BMI – biggest reason increasing risk
2. Age – second strongest factor
3. High blood pressure – increases risk
4. Physical activity – helps reduce risk

Group Insight



- Health Status: 65%
- Lifestyle & Healthcare Access: 25%
- Demographic: 10%

Health Status is the main category driving this patient's risk

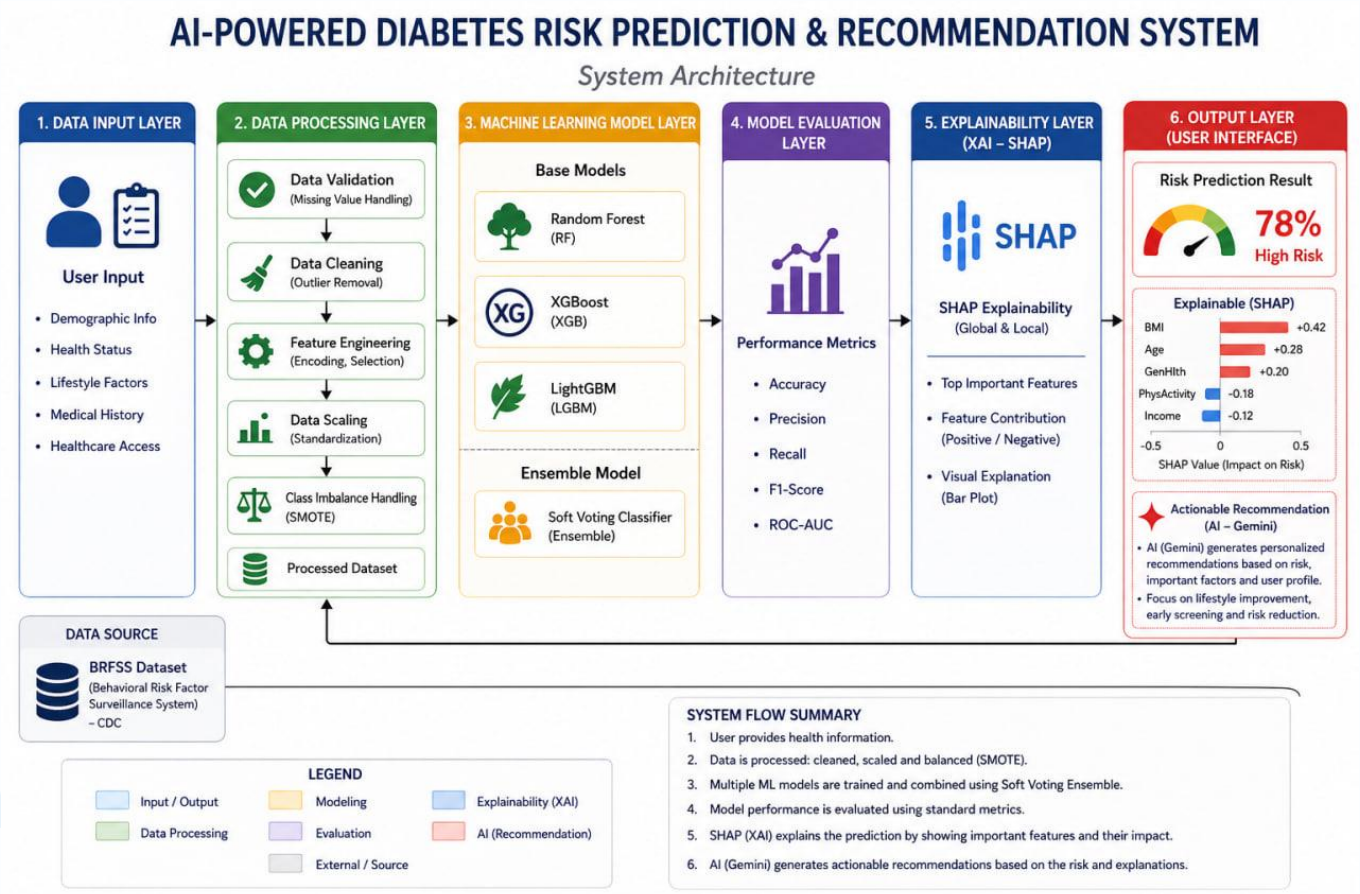
Actionable Recommendation

1. Increase regular physical activity
2. Monitor and reduce BMI
3. Check blood pressure regularly
4. Consult a healthcare professional

For education and early awareness only not a medical diagnosis

This final output extends binary prediction into probability, risk level, explanation, and action

System Architecture



Adding 2017 & 2019 BRFSS Data

2015 → 2017 → 2019



Combined data
769K rows

Diabetes Target Mapping

Raw value	Meaning	Action
1	Yes, diabetes	1 = Positive risk class
4	Prediabetes / borderline diabetes	1 = Positive risk class
3	No diabetes	0 = Negative class
2	Diabetes only during pregnancy	Remove
7 / 9 / Missing	Don’t know / Refused / Missing	Remove

Final target definition

0 = No diabetes
1 = Diabetes-related risk group
including prediabetes and diabetes

From Raw BRFSS Data to Clean Binary Data

Data Source

CDC BRFSS Annual Survey Data

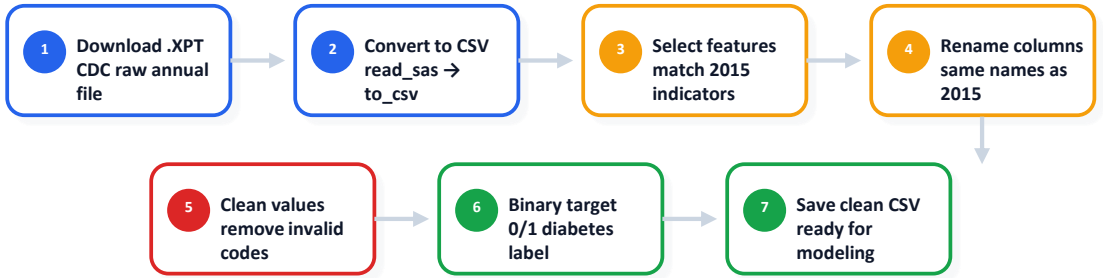
Download original yearly survey data as SAS Transport (.XPT) files.

Example years used:

2017 BRFSS raw file
2019 BRFSS raw file
2015 Kaggle clean reference

Source:
CDC BRFSS Annual Data
2017 2017 BRFSS Data (SAS Transport Format) [ZIP – 101 MB]
2019 2019 BRFSS Data (SAS Transport Format) [ZIP – 101 MB]
Kaggle Diabetes Health Indicators dataset

Cleaning Pipeline



Feature Match

Keep only columns that also exist in the 2015 diabetes dataset.

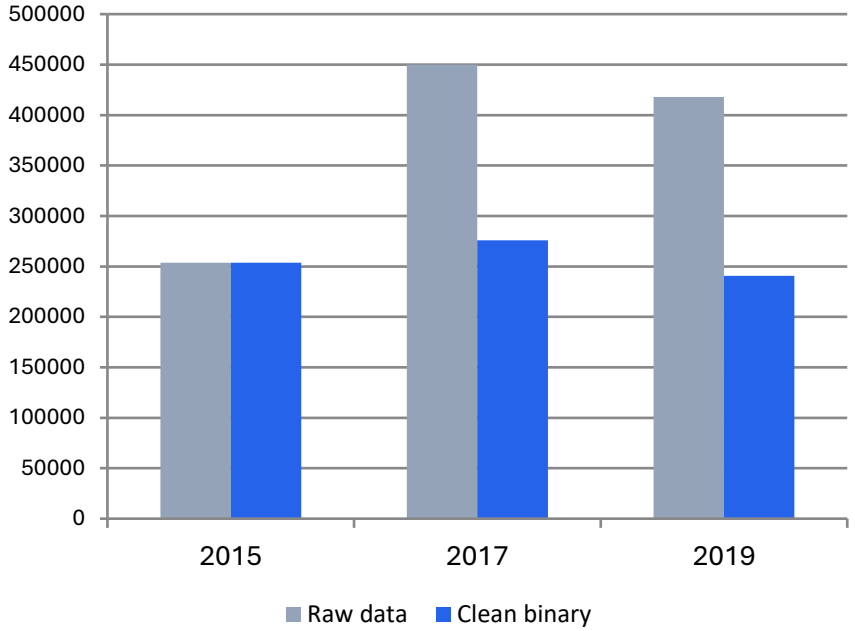
Column Names

Rename raw BRFSS codes to clear ML names such as BMI, Age, HighBP.

Binary Target

Group the diabetes column into only 2 classes: 0 and 1.

From Raw BRFSS Data to Clean Binary Data



Why rows decrease after cleaning?

Raw BRFSS has more answer categories and many survey variables. We keep only records with a clear diabetes status and the same selected 21 features used by the 2015 dataset.

253K	275K	240K
Clean 2015	Clean 2017	Clean 2019

Clean binary data = ready structure for model training

Combined Dataset and Class Imbalance



84.55%

Class 0 = 650,997
No diabetes

15.45%

Class 1 = 119,922
Prediabetes or diabetes

Total combined rows

769,919

Most records are class 0. If we train directly without care, the model may become good at predicting “no diabetes” but weaker at finding real risk cases.

Next: use stratified split + class balancing such as SMOTE